Functions, Linear Equations & *Systems*

Name: ____________________ **Date:** ___________ **Period:** ___________

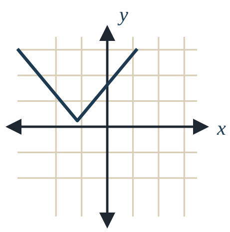
25 problems covering Sections 6.1, 6.2, 6.3, and 6.7.

Quick Reference — Unit 6 Key Formulas & Concepts

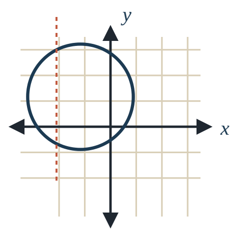
Function (6.1)

A relation in which every **input (x)** has exactly **one output (y)**.

Vertical Line Test: if any vertical line crosses the graph more than once, it's *not* a function.



✓ Function



✗ Not a function

Function Notation (6.1)

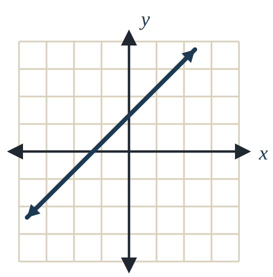
`f(x) = 2x + 4` reads "*f of x*".

To **evaluate**, substitute the input for x using parentheses.
Example: $f(3) = 2(\mathbf{3}) + 4 = \mathbf{10}$.

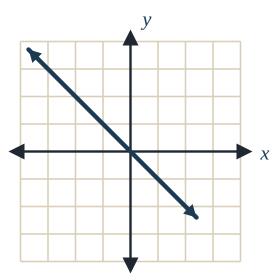
Domain = all x-values (inputs)

Range = all y-values (outputs)

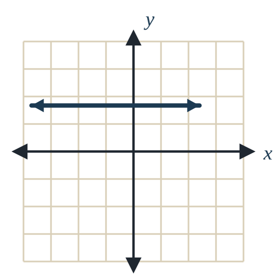
Four Types of Slope (6.2)



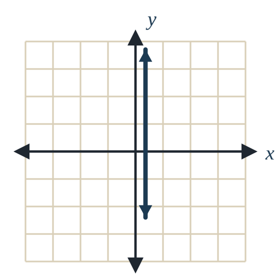
Positive
Up left to right
$m > 0$



Negative
Down left to right
$m < 0$



Zero
Horizontal
$m = 0$



Undefined
Vertical
m = undefined

Slope-Intercept Form (6.2)

$$y = mx + b$$

m = slope (rate of change; the number multiplied by x)
b = y-intercept (where the line crosses the y-axis; constant)

Slope formula: $m = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{\text{rise}}{\text{run}}$

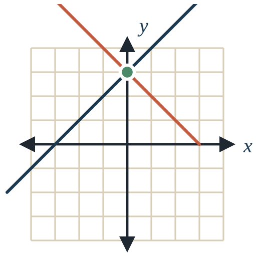
Table → Equation (6.3)

Linear if `Δy ÷ Δx` is the *same* between every pair of rows.

Steps: (1) Find slope $m = (y_2 - y_1) \div (x_2 - x_1)$. (2) Find b: use the row where x = 0, or solve y = mx + b for b. (3) Write y = mx + b.

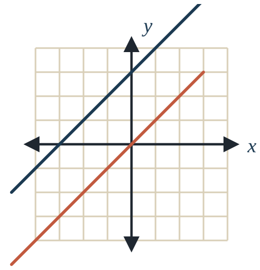
Systems of Equations (6.7)

A **solution** is the (x, y) point that makes *both* equations true — graphically, where the lines intersect.



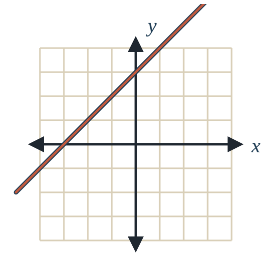
One Solution

Different slopes



No Solution

Same slope, different b



Infinite Solutions

Same slope, same b

Substitution: Solve one equation for a variable, substitute that expression into the other equation, solve, then plug back in to find the second variable.

6.1 Introduction to Functions

1. VOCABULARY

Which statement **best defines** a function?

A. A function is any equation that contains an x and a y.

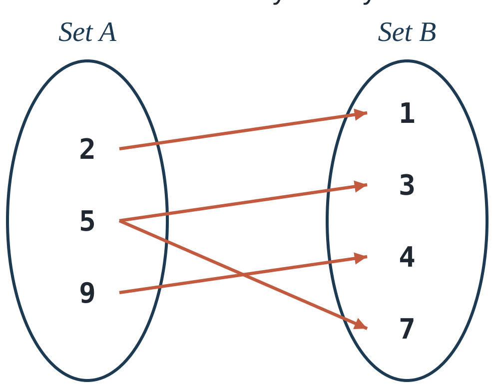
B. A function is a relation in which every input has exactly one output.

C. A function is a relation in which every output has exactly one input.

D. A function is the graph of a straight line.

2. MAPPING DIAGRAM

Does the mapping diagram below represent a function? Why or why not?



A. Yes — every input is paired with an output.

B. No — the input `5` is paired with two different outputs (`3` and `7`).

C. Yes — no two outputs are repeated.

D. No — the output `7` is paired with only one input.

3. `ORDERED PAIRS`

Which set of ordered pairs does *NOT* represent a function?

A. `{(1, 2), (3, 4), (5, 6), (7, 8)}`

B. `{(−2, 5), (0, 5), (4, 5), (8, 5)}`

C. `{(3, 1), (3, 2), (5, 4), (6, 7)}`

D. `{(−1, 0), (0, −1), (1, 1), (2, 3)}`

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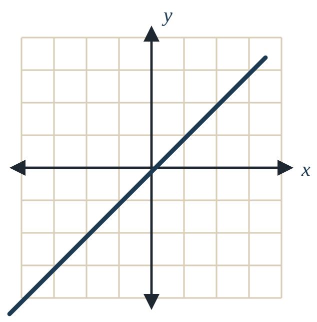
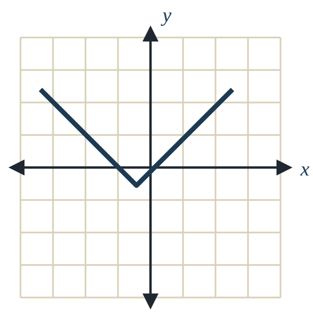
4. `VERTICAL LINE TEST`

Which graph does *NOT* represent a function? (Apply the vertical line test.)

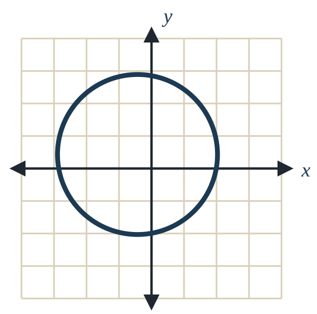
A



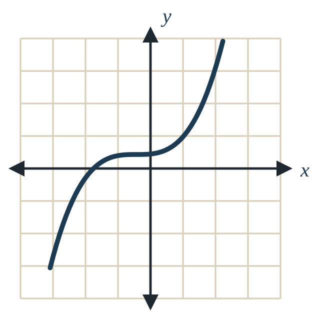
B



C



D



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5. `FUNCTION NOTATION`

Given `f(x) = 2x − 7`, find `f(−3)`.

`f(−3) =` ______________________

Enter the number only, e.g. `-13`

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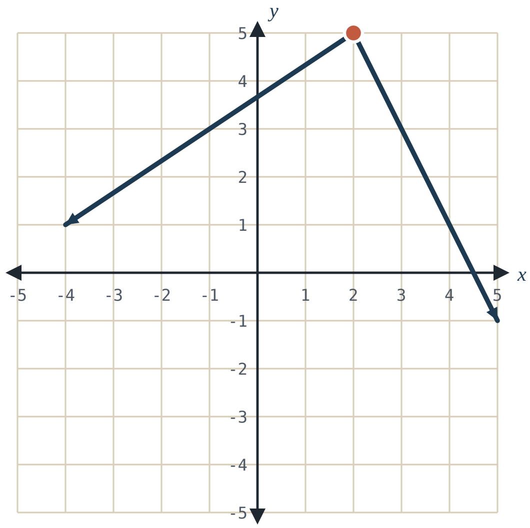
6. `EVALUATE FROM GRAPH`

Use the graph of `y = f(x)` shown below to find `f(2)`.



f(2) = ______________________________

Enter the y-value, e.g. `5`

7. `DOMAIN & RANGE`

Find the **domain** and **range** of the function defined by the table below. Express each as a set in `{ }` braces.

x	−3	0	5	8
y	4	−2	7	−2

Domain = ______________________________

Range = ______________________________

Use { } and commas — order doesn't matter, e.g. `{-3, 0, 5, 8}`

6.2 Linear Functions & Slope

8. VOCABULARY

For each line description, identify the type of slope.

A horizontal line ____________________

A vertical line ____________________

A line going up from left to right ____________________

A line going down from left to right ____________________

Choose from: Positive · Negative · Zero · Undefined

9. SLOPE FROM TWO POINTS

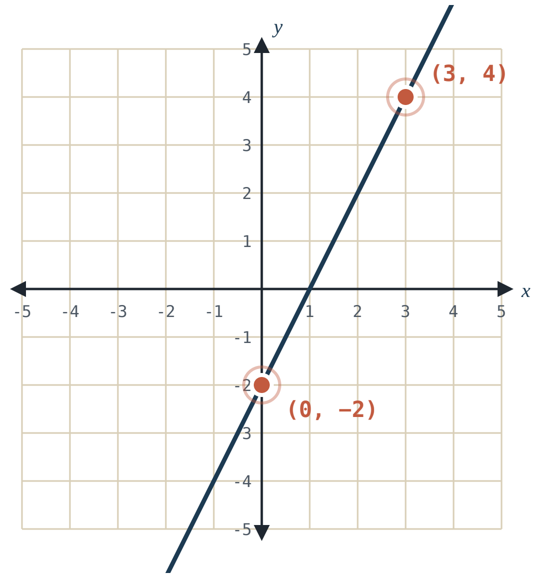
Find the slope of the line through `(−2, 5)` and `(4, −1)`. Write your answer in simplest form.

m = ____________________

Enter as a number or fraction, e.g. `-1` *or* `2/3`

10. SLOPE FROM GRAPH

Find the slope of the line shown on the graph below. Two points on the line are marked for you.



m = ____________________

Enter as a number or fraction, e.g. `2`

11. IDENTIFY SLOPE

Find the **slope** of the line: $y = -\frac{3}{4}x + 5$

m = ________________________________

Enter as a fraction, e.g. `-3/4`

12. IDENTIFY Y-INTERCEPT

Find the **y-intercept** of the line: $y = 2x - 9$

b = ________________________________

Enter the number only, e.g. `-9`

13. LINEAR OR NOT

Which table represents a **linear function**?

A

x	0	1	2	3
y	1	2	4	8

B

x	-2	-1	0	1
y	4	1	0	1

C

x	0	1	2	3
y	7	4	1	-2

D

x	0	1	2	3
y	0	1	4	9

14. ERROR ANALYSIS

A student tried to find the slope of the line through `(2, −3)` and `(8, 9)`. Their work is shown below. Identify the error and find the **correct slope**.

STUDENT'S WORK

```
m = (8 − 2) ÷ (9 − (−3))
m = 6 ÷ 12
m = 1/2
```

Correct m = ________________________________

Enter as a number or fraction, e.g. `2`

15. SOLUTIONS TO A LINEAR EQUATION

Which of the following is a **solution** to the linear equation `y = −2x + 5`?

A. `(1, 2)`
B. `(3, −1)`
C. `(0, −5)`
D. `(−2, 1)`

6.3 Writing Equations from Tables

16. MISSING VALUE

Assuming a linear relationship, find the **missing value** in the table below. *(Use the Δx and Δy columns to find the pattern.)*

x	y	Δx	Δy
1	7	—	—
2	11		
3	15		
4	19		
5	?		

y = ________________________________

Enter a number, e.g. `23`

17. IDENTIFY LINEAR TABLE

Which table represents a **linear function**?

A

x	1	2	3	4
y	2	6	12	20

B

x	-3	-1	1	3
y	8	4	0	-4

C

x	0	1	2	3
y	5	6	8	11

D

x	0	2	4	6
y	1	3	7	13

18. EQUATION FROM TABLE (X = 0)

Find the equation of the linear function in **slope-intercept form**. *(Use the Δx and Δy columns to find the slope.)*

x	*y*	*Δx*	*Δy*
0	-5	—	—
1	-2		
2	1		
3	4		
4	7		

y = ______________________________

Format: `y = mx + b` *(e.g.,* `y = 3x - 5` *)*

19. EQUATION FROM TABLE (NO X = 0)

Find the equation of the linear function in **slope-intercept form**. Note: the table does not include the x = 0 row. *(Find slope first, then solve for b.)*

x	*y*	*Δx*	*Δy*
1	1	—	—
2	-3		
3	-7		
4	-11		

y = ______________________________

Format: `y = mx + b` *(e.g.,* `y = -4x + 5` *)*

20. ERROR ANALYSIS

A student wrote an equation for the linear table below. Identify the error and write the **correct equation** in slope-intercept form.

x	y	Δx	Δy
0	4	—	—
1	7		
2	10		
3	13		

STUDENT'S WORK

Student's answer: $y = 4x + 3$

Correct equation: $y =$ ______________________________

Format: $y = mx + b$

6.7 Systems of Equations

21. VOCABULARY

A **solution** to a system of two linear equations is best described as…

A. any point that lies on either line.

B. the point where both lines have the same slope.

C. the (x, y) coordinate that makes *both* equations true.

D. the x-value where one line equals zero.

22. SOLUTIONS BY INSPECTION

How many solutions does this system have? $y = 2x + 5$ $y = 2x - 1$

A. One solution

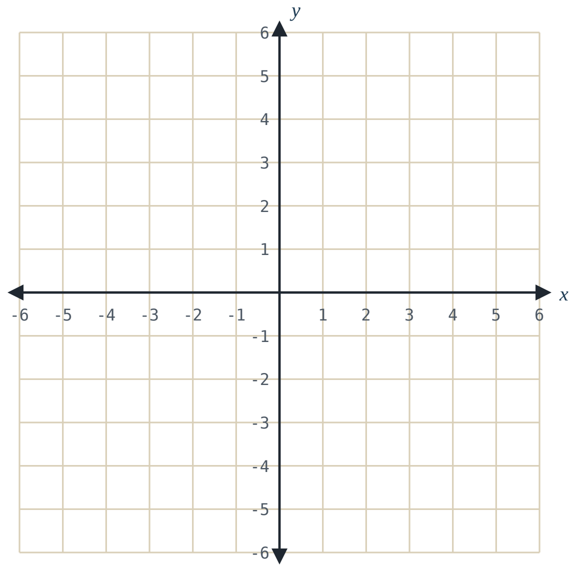
B. No solution

C. Infinite solutions

D. Cannot be determined

23. SOLVE BY GRAPHING

Graph the following system on the grid below and find the solution. *(Each line's y-intercept appears on the grid — start there, then use the slope.)*



$y = 2x - 3$

$y = -x + 3$

x = ______________________

y = ______________________

Enter integers, e.g. 2

24. SUBSTITUTION

Solve the system by substitution. $y = 2x + 1$ $y = -x + 7$

x = ______________________

y = ______________________

Enter integers, e.g. 2

25. SUBSTITUTION

Solve the system by substitution. $y = x - 4$ $3x + 2y = 7$

$x =$ ____________________

$y =$ ____________________

Enter integers, e.g. `3` *or* `-1`